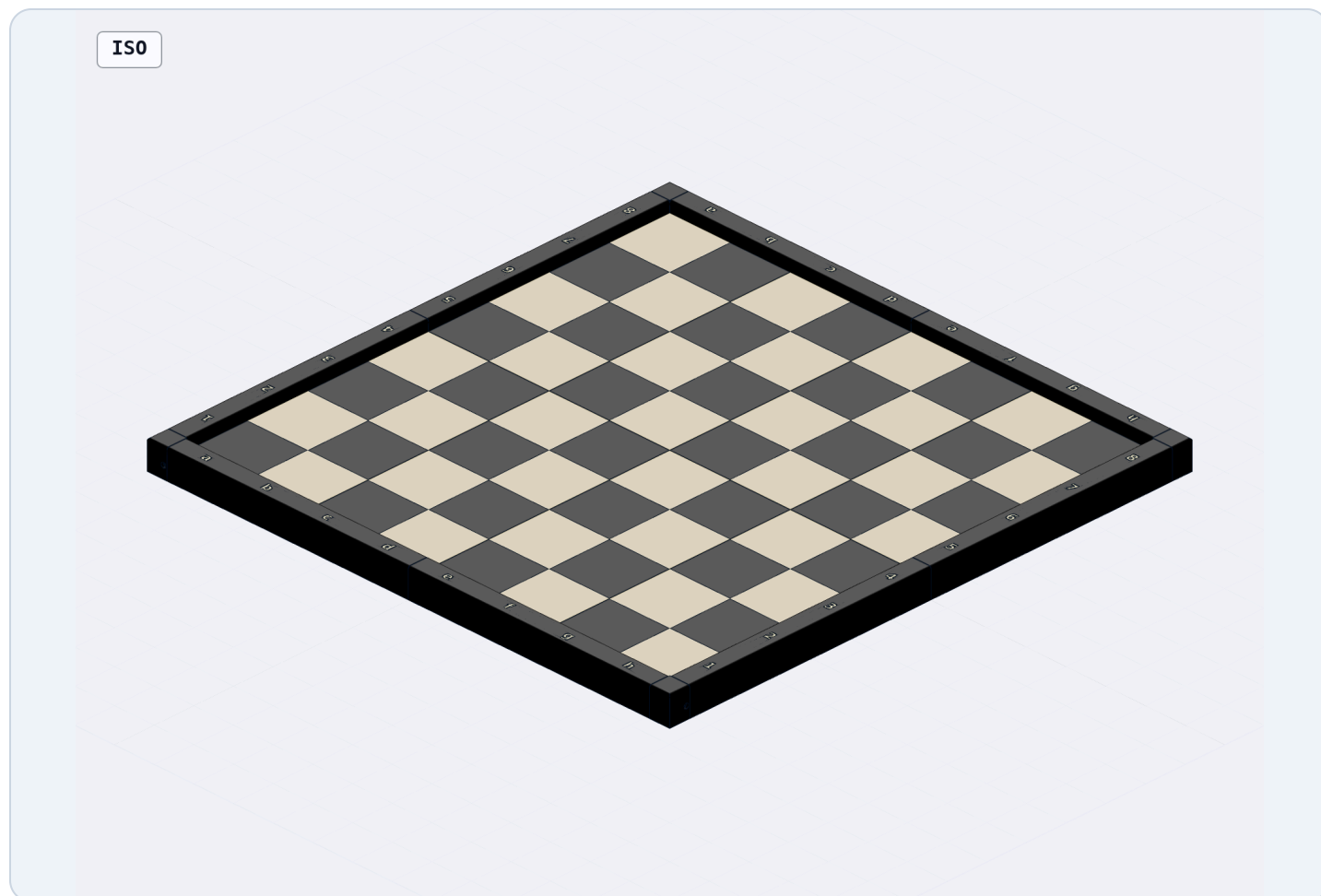


# 60 mm Modular Chessboard

Eight-panel, full-size board for the Bambu Lab X2D, with raised perimeter, contrasting algebraic notation, concealed seam registration, and 45-degree mortise-and-tenon corner joints under screw-retained caps.

**60 mm**

square size

**480 mm**

playing area

**520 mm**

overall size

**10 mm**

perimeter rise

**Read before assembly.** Several nuts become inaccessible after adjoining parts are installed. Follow the order in this manual and complete every “load nut first” step before closing a seam or adding a corner cap.

Designed for PLA Matte. Hardware: M3×12 and M2×12 screws with standard hex nuts. No heat-set inserts are required.

# Parts, hardware, and preparation

## Printed parts

| PART                                      | QTY. |
|-------------------------------------------|------|
| Named 1/8 playing-panel bodies            | 8    |
| Named perimeter rails                     | 8    |
| Underside seam bridges                    | 16   |
| Corner caps                               | 4    |
| Glue-in dark-square inlays                | 32   |
| Notation insert set, separate-print route | 1    |

## Hardware

| HARDWARE                  | QTY. |
|---------------------------|------|
| M3×12 screws              | 48   |
| Standard M3 hex nuts      | 48   |
| M2×12 screws              | 4    |
| Standard M2 hex nuts      | 4    |
| Adhesive felt/rubber pads | 12   |

Nominal nuts: M3 = 5.5 mm across flats × 2.4 mm thick; M2 = 4.0 mm across flats × 1.6 mm thick.

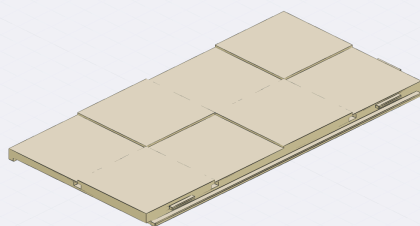
### Useful tools

- Drivers matching your M3 and M2 screw heads
- Tweezers or needle-nose pliers for nut placement
- Deburring tool, hobby knife, or small flat file
- Protected, flat work surface and soft cloth
- PLA-compatible adhesive for separate inlays

### Identify the faces

- **Playing side:** checker pattern and raised border
- **Underside:** screw recesses, nut entrances, bridges
- **Internal edges:** lower tongue/groove plus upper registration keys
- **Outer edges:** grooves that receive rail tongues
- **Panel map:** south/north row plus file pair ab, cd, ef, or gh

ISO



### Color parts

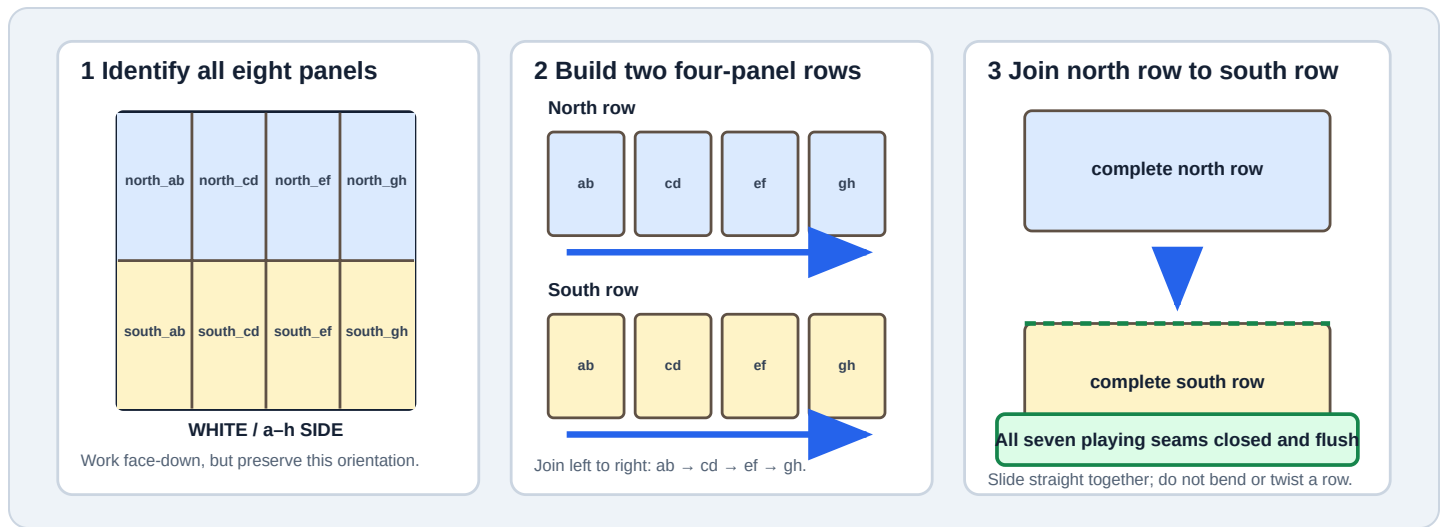
Print each **panel\_\*\_light\_body** in the chosen light color. Print the 32 **dark\_square\_inlay** parts on the two dedicated dark-color plates. Dry-fit every square, then apply a thin, even layer of PLA-compatible adhesive and press it fully into its recess until flush. Keep each rail as one object and assign its inset notation a contrasting filament.

### Clean fit surfaces

Remove brim material, strings, and burrs. Preserve the 0.5 mm lead-ins and 0.4 mm first-layer relief. Never force a connector that has visible debris.

**Hardware trial:** Test one M3 nut and one M2 nut before loading all pockets. On the 1/8 panels, the M3 nut passes through a 6.1 mm entry, seats in a 5.8 mm across-flats pocket, and aligns with a 3.6 mm screw passage.

# Join the eight playing panels



Build the two four-panel rows first, then slide the complete north row onto the south row.

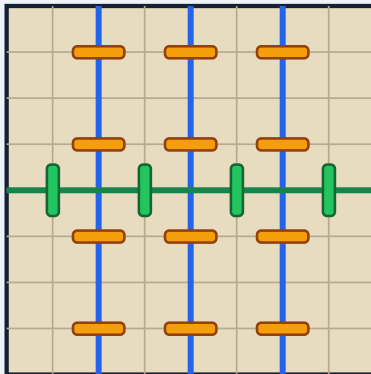
- 1 Work face-down.** Lay out the south row as **south\_ab, south\_cd, south\_ef, south\_gh**; place the matching north row above it. White's a–h side is at the south/bottom.
- 2 Load all M3 nuts before joining.** Every panel has six entrances: two on each long edge and one on each short edge. Seat all 48 nuts over their vertical passages and test-thread each location.
- 3 Join the south row.** Work left to right: ab → cd → ef → gh. Keep the panels supported and move each female groove straight onto the preceding panel's tongue and upper keys.
- 4 Join the north row.** Repeat left to right with north\_ab through north\_gh.
- 5 Join the two rows.** Slide the completed north row southward onto the north-facing tongues of the south row. The upper keys register the playing faces while the lower tongues carry shear.
- 6 Check all seven playing seams.** Sight across the playing face while supporting the full assembly. All eight panels should be level with no trapped filament or visible step.

**Designed clearance:** the lower tongue/groove and concealed upper keys use 0.25 mm nominal depth clearance. A smooth sliding fit is expected; a press fit is not.

**Do not force the joint.** If the panels stop before a seam closes, separate them and remove brim, stringing, or debris from both profiles.

# Lock all seven panel seams from below

Underside bridge plan — 16 bridges total



WHITE / a-h SIDE

12

## Across the three vertical panel seams

Four bridges per seam at  $Y = -180, -60, 60, \text{ and } 180 \text{ mm}$ .  
Bridge bodies run horizontally, crossing each blue seam.

4

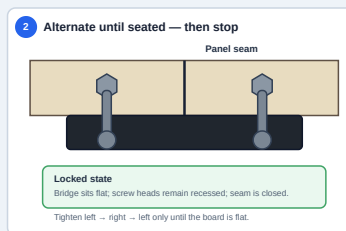
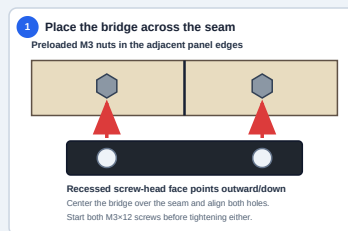
## Across the horizontal center seam

One bridge at  $X = -180, -60, 60, \text{ and } 180 \text{ mm}$ .  
Rotate these bridge bodies  $90^\circ$  so they cross the green seam.

### Install while the board is face-down and fully supported.

Each bridge uses two M3×12 screws: one into the captured nut on each side of its seam.  
Start both screws, alternate tightening, and stop as soon as the bridge sits flat.

Underside seam bridge — both screws pull against captured M3 nuts



### Three vertical seams

Install four bridges on each seam at  $X = -120, 0, \text{ and } 120 \text{ mm}$ : twelve bridges total.

### One horizontal center seam

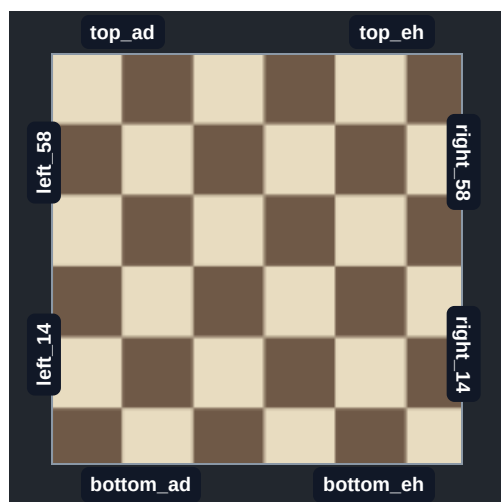
Install four bridges at  $X = -180, -60, 60, \text{ and } 180 \text{ mm}$ , rotated  $90^\circ$ .

- 1 **Keep the board face-down and fully supported.** Confirm the correct M3 nuts were loaded before the seams were closed.
- 2 Place each bridge across its seam with the two recessed screw-head seats facing outward/downward.
- 3 Insert two M3×12 screws through the bridge and into the captured nuts. Turn each screw only a few threads at first.
- 4 Alternate between the two screws until the bridge sits flat. Repeat for all sixteen bridge locations.
- 5 Sight across the playing side after supporting the full board. Loosen and reseal any bridge associated with a visible seam lip.

**How the bridge locks:** each screw pulls against a nut captured inside one of the two adjacent panel edges. The bridge clamps the seam closed and resists bending without exposed playing-side hardware.

**Stop when seated.** Excess preload can bow the 240 mm panels, crush printed layers, or cause long-term PLA creep.

# Install the raised perimeter rails



White sits at the bottom a–h side. The a1 square must be dark.

## Rail positions

|                  |                      |
|------------------|----------------------|
| White side       | bottom_ad, bottom_eh |
| Opposite side    | top_ad, top_eh       |
| Left from White  | left_14, left_58     |
| Right from White | right_14, right_58   |

## Notation check

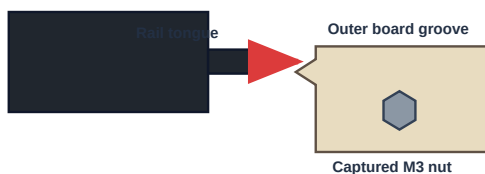
Letters face their adjacent player. Rank numerals face the player for whom that rail is on the right: right/east reads upright from White; left/west reads upright from Black. The two number sets are 180° opposed.

- 1 **Install the mortise-side rails first:** left\_14, bottom\_eh, right\_58, and top\_ad. Push each long tongue straight inward into the panel groove, then fasten it with two M3×12 screws.
- 2 Load one M2 nut into each locking rail before installation: bottom\_ad, right\_14, top\_eh, and left\_58.
- 3 Hold each locking rail at its final longitudinal position. Push it straight inward so its long rail tongue enters the panel groove while its low corner tenon enters the perpendicular rail's open mortise.
- 4 Install two M3×12 screws in each locking rail. Confirm both 45-degree corner faces seat evenly, then leave the caps off for the next stage.

## Perimeter rail — slide the tongue into the board, then fasten from below

### 1 Engage the full-length rail tongue

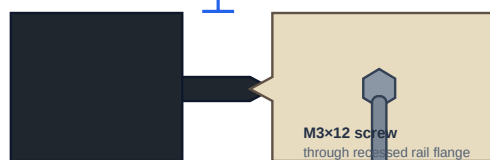
Black raised rail



Slide the rail straight along the entire edge.  
Do not use the screws to pull a blocked tongue into place.  
The 0.5 mm lead-in should start the connection easily.

### 2 Fasten the seated rail from below

10 mm rise



**Correct locked state**

Tongue fully seated; border continuous.

Start both sets of screws before tightening.

## Correct fit

The rail tongue and corner tenon seat together, the two 45-degree flange faces do not overlap, and the perimeter top remains 10 mm above the playing surface.

## If a rail stops early

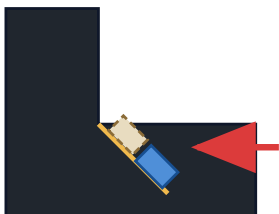
Remove it and inspect the panel groove, tenon, and mortise for brim or stringing. Push the rail straight inward; do not slide it toward the corner or draw it home with screws.

**Maximum printed length:** each rail is 247.05 mm long. If reprinting, position a 3–4 mm brim carefully inside the X2D's 256 mm main-nozzle area.

# Seat the miter joints and lock the caps

45° corner lock — mortise rail first, tenon rail inward, cap and cross-screw last

## 1 Engage the tenon



**Push the locking rail straight inward**  
Its panel tongue and corner tenon seat together.  
Yellow line = complementary 45° miter faces.

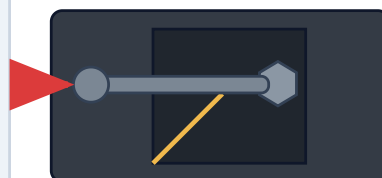
## 2 Load nut; lower cap



**Cap slots capture both mitered cores**  
Keep the cap level and its recessed hole facing the named locking rail.

## 3 Install the cross-screw

Tighten only until the cap is retained



### Locked state

Tenon + cap + M2×12 retain the corner.

Do not use the screw to pull an unseated miter closed.

## Load one M2 nut at each corner

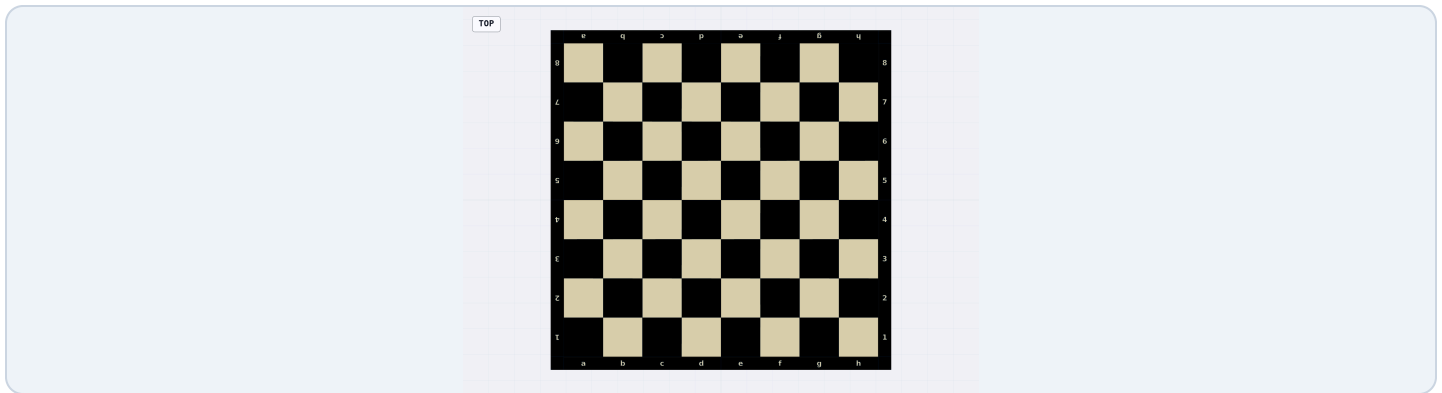
| CORNER    | LOCKING CORE THAT RECEIVES THE NUT | CAP SCREW DIRECTION          |
|-----------|------------------------------------|------------------------------|
| Southwest | bottom_ad                          | From the nearby outside face |
| Southeast | right_14                           | From the nearby outside face |
| Northeast | top_eh                             | From the nearby outside face |
| Northwest | left_58                            | From the nearby outside face |

- 1 Confirm the locking rail's tenon is fully inside the perpendicular rail's mortise and the two 45-degree core faces are evenly separated. The underside flange faces must not overlap.
- 2 Confirm the M2 nut lies flat in the hex pocket and its threaded hole aligns with the horizontal passage through the locking core.
- 3 Rotate the identical corner cap so its recessed cross-hole faces the selected core and is accessible from the nearby outside edge.
- 4 Slide the cap down over both cores. The shallow slot lead-ins help both enter together; keep the cap level.
- 5 Insert one M2×12 screw through the cap wall and locking core, then thread it into the captured nut.
- 6 Tighten only until the screw head sits in the 4.5 mm diameter × 2.2 mm deep recess and the cap cannot lift.
- 7 Repeat at all four corners.

**How the lock works:** the low mortise-and-tenon resists relative corner movement, the 45-degree faces remove the former flange collision, and the cap captures both cores. The cross-screw positively prevents the cap from lifting off.

**Do not substitute a longer screw without checking.** M2×12 is the validated length. A longer screw may bottom out or damage the printed pocket.

# Final setup, inspection, and care



## Final checklist

- |                                                                                     |                                                                        |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <input type="checkbox"/> All 16 seam bridges installed with 32 M3 screws            | <input type="checkbox"/> All 8 rails installed with 16 M3 screws       |
| <input type="checkbox"/> All 4 caps locked with M2×12 screws                        | <input type="checkbox"/> No screw head protrudes below a recessed seat |
| <input type="checkbox"/> Playing-surface seams are flush                            | <input type="checkbox"/> a1 is dark and notation is correctly oriented |
| <input type="checkbox"/> Tenons are seated and all 45-degree corner joints are even | <input type="checkbox"/> No loose nuts rattle inside the assembly      |

- 1 Apply eight felt/rubber pads to the rail recesses and four to the corner-cap recesses.
- 2 With assistance if needed, support the full 520 mm board and turn it playing-side up. Do not lift it by a single rail or corner cap.
- 3 Place it on a flat surface and check for rocking. If needed, verify pad thickness and loosen/reseat a bridge before retightening evenly.
- 4 After the first several uses, check that the screws remain snug. PLA does not benefit from high fastener preload.

### Tight connector

Remove the part, clean the entrance, and lightly deburr only the interfering edge. Do not hammer the panels, rails, or caps together. If consistently tight after reprinting, use small slicer XY-hole/contour compensation.

### Visible seam lip

Check for debris in the upper-key grooves, verify the assembly is on a flat surface, and tighten opposing bridge screws gradually. The keys register height; the bridges establish final stiffness.

### Cap screw will not start

Remove the cap and confirm it was rotated toward the selected locking core. Reseat the M2 nut flat in its top-loaded hex pocket and try again without forcing the thread.

### Routine care

Clean with a dry or lightly damp cloth. Avoid hot cars, heaters, direct prolonged sun, solvents, and soaking. Store the eight playing panels supported flat to minimize PLA creep.

**Design reference:** STEP/STL files and the latest validation report are in the project folder. The generated assembly measures 520 × 520 × 27.6 mm and contains eight playing panels, sixteen seam bridges, and 60 mm squares over a 480 × 480 mm playing area.